When Should a Sandbox Exist?

Acceptance-Aware Speculation Without False Confidence

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Abstract

An executable fork makes alternative futures cheap, but it does not manufacture independent evidence or grant a child the authority to accept its own work. We formulate a *sandbox existence test* (fidelity, private divergence, authority, and economics), then borrow the useful part of speculative decoding: calibrated acceptance predictions and adaptive allocation of a finite verification budget. Exact token-sampling acceptance is distinguished from the probability that a software artifact passes a trusted verifier. Conditional acceptance handles correlated failures; lineage-aware reduction and selection-aware inference prevent false precision after choosing winners. We give a minimal controller protocol and a falsifiable experiment compatible with the Cambridge seminar's existing preregistration. All new scheduler, covariance, and selection-aware components are **design proposals**, not measured runtime gains.

1 First ask whether a fork is real, useful, and safe

A sandbox is an *execution/policy boundary*, not a synonym for a microVM or a process. A *forkable continuation* is a private branch of a declared reached state S_0. Its contents may include filesystem, environment and optionally live memory/processes; copied guest state does *not* copy the external world or publication authority. The Cambridge talk distinguishes named-snapshot *restore fan-out* from live-memory fork until the nonce/process-survival probe passes [1].

For a task class $\mathcal{T}$, the following are gates, not performance results:

Decision rule. Call it a *forkable sandbox* only when the state semantics and isolation/authority gates hold. Call it an *advantageous* fork only when the same-workload warm alternative loses after all overheads. Performance is not a security definition.

For K equivalent tasks, a simple *work-cost* accounting is

$$\begin{aligned} C_{\text{warm}} &= K(P_w + E + V), \\ C_{\text{fork}} &= P_0 + H + K(R + E + V) + D. \end{aligned} \tag{1}$$

Here P_w is warm preparation, P_0 preparation of the captured parent, H capture, R restore, E per-trial execution, V exact-artifact verification, and D divergence/coordination. Under this simplified additive model, fork saves work iff

$$\boxed{KP_w > P_0 + H + KR + D.} \tag{2}$$

Gate	Required evidence before claiming a useful fork
State	Name the bytes, runtime state and external dependencies needed by the next action; say explicitly whether memory survives.
Fidelity	On pinned tests $t \in \mathcal{T}$, parent and restored continuation agree on required observations under controlled inputs; record failures.
Privacy	Each child gets private writes and unique runtime identity/RNG; no child can modify another's evidence or the verifier.
Authority	Controller owns immutable check definitions, epoch, evidence ledger and promote permission; secrets/leases are reissued, never trusted because they were copied.
Effects	Network/API/billing consequences are fenced, simulated, or idempotently reconciled; restore never undoes outside effects.
Cost	Compare against an equivalent *warm cached* baseline, including capture, restore, divergence, verification and contention.

Table 1: Operational criteria; none follows from a fast create benchmark alone.

Latency additionally depends on critical path, contention, concurrency, and tail behavior; hedged requests are a separate tail-latency precedent [7]. Cheap restore alone never proves lower end-to-end latency. This matches the seminar's preregistered H1 comparison to a *warm cache*, not merely a cold start [1].

2 What speculative decoding really proves

Let $q(x)$ be a cheap proposal distribution and $p(x)$ the exact target distribution over tokens. Draw $x \sim q$ and accept with probability

$$a(x) = \min\left(1, \frac{p(x)}{q(x)}\right), \qquad q(x) > 0. \tag{3}$$

Averaging over proposals gives *maximal-coupling* acceptance:

$$\begin{aligned} \alpha_{\text{token}} &= \sum_x q(x)a(x) = \sum_x \min\{p(x), q(x)\} \\ &= 1 - \underbrace{\frac{1}{2}\sum_x |p(x) - q(x)|}_{\text{TV}(p,q)}. \end{aligned} \tag{4}$$

Upon rejection, sample from $r(x) = [p(x) - q(x)]_+ / \mathrm{TV}(p, q)$ when $\mathrm{TV} > 0$. Then

$$\min(p, q) + \mathrm{TV}(p, q) r = p, \tag{5}$$

pointwise. Thus the emitted token retains *exactly* the target distribution; no coupling can reuse a q-sample with a higher probability of equality [3]. The theorem depends on *known* p, q and this correction rule.

Never substitute a test pass for Eq. (4). For code, A_i = {exact candidate i passes trusted checks}, and $a_i = \mathbb{P}(A_i \mid \text{history})$ is a *predictive verifier-acceptance rate*. It is generally neither $1 - \mathrm{TV}$ nor the probability of semantic correctness. A verifier can be incomplete or compromised. The controller must own it.

2.1 A precise, but narrow, Gaussian bridge

For two normalized Gaussians $p = N(\mu_1, \Sigma)$ and $q = N(\mu_2, \Sigma)$ with the *same* positive definite covariance, the paper's two-factor weighted disagreement is

$$\Delta = \tfrac{1}{2}(\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 - \mu_2). \tag{6}$$

Their TV overlap and KL divergence obey

$$\boxed{\alpha_{\text{token}} = 2\Phi\left(-\sqrt{\Delta/2}\right), \quad D_{\mathrm{KL}}(p \| q) = \Delta.} \tag{7}$$

This is an *identity for these two distributions*, not a general map from worker disagreement to patch correctness. Identical Gaussian means with unequal covariances give $\Delta = 0$ but nonzero TV. Identical wrong patches can have perfect apparent agreement without any new evidence [2].

3 Acceptance-aware fork selection

Let A_i denote verifier acceptance and H encode fork ancestors, model/prompt/test hashes, failures and known shared inputs. In a fixed trial order, define

$$a_i^- = \mathbb{P}(A_i \mid H, A_1^c, \ldots, A_{i-1}^c). \tag{8}$$

The chain rule gives an *exact* correlated-success identity:

$$\boxed{\mathbb{P}\left(\bigcup_{i=1}^{K} A_i \,\middle|\, H\right) = 1 - \prod_{i=1}^{K} (1 - a_i^-).} \tag{9}$$

Replacing a_i^- by marginal success rates silently assumes independence. The predictors must be conditioned on negative evidence from failed siblings.

For a fixed set S of candidate branches let $F(S) = \mathbb{P}(\bigcup_{i \in S} A_i)$. A new worker j contributes

$$\boxed{F(S \cup \{j\}) - F(S) = \mathbb{P}\left(A_j \cap \bigcap_{i \in S} A_i^c\right).} \tag{10}$$

Diversity is an objective, not decoration. What matters is the probability that j succeeds *when the current workers fail*. Since the event on the right shrinks as S grows, F is monotone and submodular for a *fixed* joint distribution. Static equal-cost greedy selection has the classic $(1 - 1/e)$ approximation to the optimal fixed-size set. This guarantee does *not* transfer automatically to a history-dependent adaptive policy, whose conditions require separate proof [6].

3.1 Concrete correlation penalty

A transparent illustration takes a shared latent difficulty $Z \sim \mathrm{Beta}(u, v)$ and conditionally IID branch outcomes $A_i \mid Z \sim \mathrm{Bernoulli}(Z)$. Then

$$\begin{aligned} \mu = \frac{u}{u+v}, \quad \rho = \frac{1}{u+v+1}, \\ \mathbb{P}(\exists \text{ pass}) = 1 - \frac{(v)_K}{(u+v)_K}. \end{aligned} \tag{11}$$

With $(u, v) = (0.45, 1.05)$ and $K = 8$, each branch has marginal pass rate 0.30 and pairwise outcome correlation 0.40. The eight-branch success probability is *about 65.8%*, versus $1 - 0.7^8 \approx 94.2\%$ under independence. After four failures, the next same-family pass probability is $u/(u + v + 4) = 8.18\%$. These are **model calculations**, not measurements of islo or the factory.

3.2 Depth, width, and verification cost

For a dependent chain of D speculative edits, let L be the number of consecutive edits whose exact resulting artifacts pass, stopping at the first failure. The tail-sum formula is

$$\mathbb{E}[L] = \sum_{j=1}^{D} \mathbb{P}(A_1 \cap \cdots \cap A_j). \tag{12}$$

If conditional prefix survival is constant a, then $\mathbb{E}[L] = \sum_{j=1}^{D} a^j$. Unlike token decoding, a rejected code patch has no free residual-correction token: repair, rollback, rerun and verifier invocations all cost resources.

A controller can choose depth D, width W, model/prompt diversity, verifier order and cancellation policy to maximize a budgeted objective such as

$$\max_{\pi} \ \mathbb{P}_\pi(\text{verified success by } T) - \lambda \mathbb{E}_\pi[C], \tag{13}$$

subject to *unrelaxed* authority and final-artifact checks. Predictive acceptance must be calibrated on held-out traces, with latency and compute jointly measured. EAGLE-2 motivates dynamic trees from context-dependent acceptance; EAGLE-3 motivates improving the drafter rather than simply widening speculation. Neither supplies a correctness theorem for arbitrary sandbox tasks [4, 5].

4 Acceptance is not independent evidence

In the paper's independent Gaussian common-target model, worker i returns estimate $\hat{\theta}_i$, information P_i, evidence IDs and lineage. The current reducer forms $P = \sum_i P_i$ and $q = \sum_i P_i \hat{\theta}_i$, rejects declared duplicate IDs and transports ancestry. It *does not* infer covariance from lineage or implement selection correction [2].

If estimates share latent error U and have independent noise ε_i, write $Y_i = \theta + U + \varepsilon_i$ with variances τ^2 and σ^2. Then

$$\mathrm{Var}(\bar{Y}) = \tau^2 + \frac{\sigma^2}{K}, \quad K_{\mathrm{eff}} = \frac{K}{1 + (K-1)\rho}, \tag{14}$$

where $\rho = \tau^2/(\tau^2+\sigma^2)$. Eight estimates with $\rho = 0.4$ have $K_{\mathrm{eff}} \approx 2.11$ under this model. *Acceptance correlation* and *estimation-error correlation* are separate quantities; this effective sample size is not a success-probability shortcut.

With a *justified* covariance matrix C and a common-target observation map H_0, generalized least squares gives

$$\begin{aligned} \hat{\theta} &= (H_0^T C^{-1} H_0)^{-1} H_0^T C^{-1} Y, \\ \mathrm{Cov}(\hat{\theta}) &= (H_0^T C^{-1} H_0)^{-1}. \end{aligned} \tag{15}$$

A candidate fork-DAG covariance model is $C_{ij} = \sum_{a \in \mathrm{Anc}(i) \cap \mathrm{Anc}(j)} T_a + \mathbf{1}_{i=j} R_i$. Ancestry alone does *not* identify these variance components: fit/validate them externally or **abstain** from claiming increased precision.

Selection creates another bias. Suppose 100 IID $N(\theta, 1)$ estimates each report $[Y_i - 1.96, Y_i + 1.96]$. If we choose the largest estimate, coverage becomes

$$\mathbb{P}(\theta \in I_{\arg\max Y_i}) = \Phi(1.96)^{100} - \Phi(-1.96)^{100} \approx 7.95\%. \tag{16}$$

So even initially valid 95% intervals may become invalid after choosing winners. A valid reducer must record *rejections and selection history*, then use simultaneous/selection-aware or held-out inference; unchanged marginal intervals are not automatically valid after search. This is a **synthetic calculation**, not measured agent behavior.

5 A minimal end-to-end controller

Admission. An immutable work-order key binds repository tree, initial-state digest, tool/model/prompt versions, task and selection context. Retried evaluation of *the same observation* reuses its evidence token; new samples get new tokens, but keep common ancestry. Do not count retries or speculative replicas twice.

Execution. Keep warm compiler/dependency state where safe; restore private writable state; reseed uniqueness and remint child authority outside the copied guest. Stop or cancel losers only when their recorded outcomes are no longer necessary. A cancellation is not a successful result.

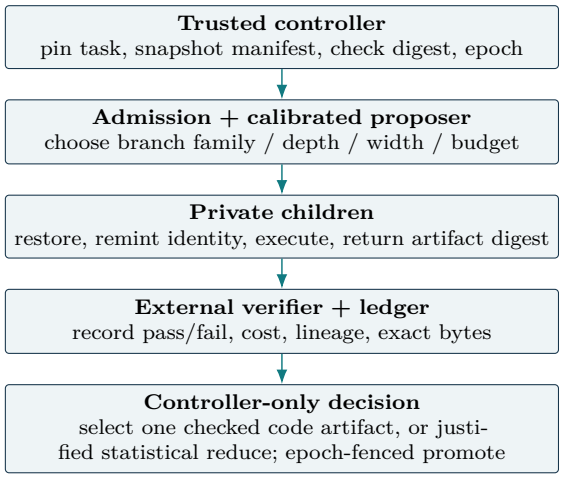


Figure 1: Proposed protocol. Only the controller can issue the final promote operation.

Verification. An external checker binds `(input tree, output digest, verifier digest, epoch)` to a signed or controller-authenticated receipt. Keep the negative outcomes, including timeouts, refusals and missing checks. Different result types have different fan-in: choose one code patch, recheck a composed patch, or pool measurements only under a common target and justified covariance.

Publication. Persist exact artifact bytes and checks before tearing down the worker. Check the current epoch/version at a single controller-owned compare-and-swap. Stale children can finish but cannot publish. An untrusted child cannot write the score, tests, receipt or publish token; network effects are fenced or reconciled. The seminar's serial factory records motivate these obligations but did not demonstrate an integrated forked version [1].

6 Falsifiable evaluation, not a speedup claim

Preserve preregistration. The Cambridge `PREREGISTRATION.md` already fixes Gate 0 (memory nonce), fidelity, H1 warm-cache cost and H2 sibling-vs-stranger correlation, with predefined tests and rejection thresholds. Do *not* change those analyses or imply they have run. The following is a separately labelled **exploratory Phase D proposal**.

1. **Same tasks, same authority:** pin repositories, work orders, checker images/digests, snapshot semantics and compute/time budgets; compare serial warm, fixed-width fan-out, calibrated acceptance-only, and lineage-aware adaptive scheduling.
2. **Predict before seeing the outcome:** train conditional pass models on disjoint historical tasks; freeze and calibrate them; record predictive log loss, Brier score and reliability by fork family, failure class and

workload difficulty.

3. **Measure complete cost:** prepare/capture/restore, agent calls, verifier executions, cancellation, external effects, queueing, memory and p50/p95 time-to-*first exact verified artifact*; count CPU-seconds and dollars per success. Include no-fork/warm-cache controls.
4. **Try hard negative controls:** cloned wrong assumptions, altered prompts/models, intentionally missing direct tests, stale publication epochs, duplicate evidence IDs, verifier-tampering attempts, unknown shared noise and post-selection intervals.
5. **Report falsifiers:** adaptive width helps only if additional verified successes per budget exceed overhead; lineage helps only if it improves calibrated conditional success or coverage on held-out tasks; refuse unjustified precision rather than reporting a fabricated gain.

Talk placement. A 90-second optional Q&A bridge after the seminar's runtime/evidence boundary: *"A fork is worthwhile only if it preserves the state we need, costs less than the warm alternative, and cannot grade or publish itself. Speculation then asks: what is the probability this* next *branch succeeds* when the existing branches fail*? That probability controls the scheduler, while independent evidence controls statistical confidence."*

Takeaway

Speculate to save latency. Verify exact outcomes. Preserve rejected attempts.
Count evidence by provenance and calibrated dependence, never by the number of sandboxes or the number of green checks.

References

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